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College of Osteopathic
Medicine

Pediatric Cardiac Considerations Part 3

CARDIOVASCULAR SYSTEM

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- Summarize the pathogenesis of **Kawasaki disease, rheumatic fever/disease, cardiomyopathies** and **syncope** in the pediatric population
- Describe the clinical presentation and management of these disease states
- Construct a working differential diagnosis for these diseases
- Be able to discuss the treatment for these disease states

SESSION OBJECTIVES

- Pediatrics for Medical Students 3rd edition, Lippincott Williams & Wilkins
- Nelson's Textbook of Pediatrics, 22nd edition, Elsevier Saunders
- Rudolph's Pediatrics, 24nd edition, McGraw Hill
- Park's Pediatric Cardiology for Practitioners, 7th ed., Elsevier
- Essentials of Pediatrics, 10th edition, Elsevier
- ScholarRx Brick on rheumatic heart disease:
<https://exchange.scholarrx.com/brick/acute-rheumatic-fever-and-rheumatic-heart-disease>

RESOURCES

Case #1

CC: 18yo teen presents with fainting on the first day of college.



HPI-

18 yo teen presents to the college health center because of a fainting spell while at school. She was rushing to an 8am class after missing her morning alarm. She got out of bed, brushed her teeth, washed her face, got dressed and ran to class.

While in her seat, she felt lightheaded, was perspiring and then fainted for approximately 1 min. Her professor intervened and the student was walked over to the health center.

PMHx: none

PSHx: none

Immunizations: UTD

Meds: none

Soc: denies drugs/tob, admits to drinking alcohol with friends the night before

Physical Exam

98°F oral, HR: 120, RR: 18 Wt: 120lbs Ht: 64" BP: 132/72 pulse ox.- 98%

General: Well appearing adolescent female in no distress but appears fatigued

HEENT: WNL

Lungs: CTA bilaterally, no WRR

Cardiac: increased rate, regular rhythm, no murmur

Abd: Soft, non distended, nontender, +BS, no HSM

Upper and Lower Extremities: pulses symm. and normal, strength 5/5 b/l, nontender to palpation, no signs of muscle atrophy

Neuro: grossly normal except for fatigue

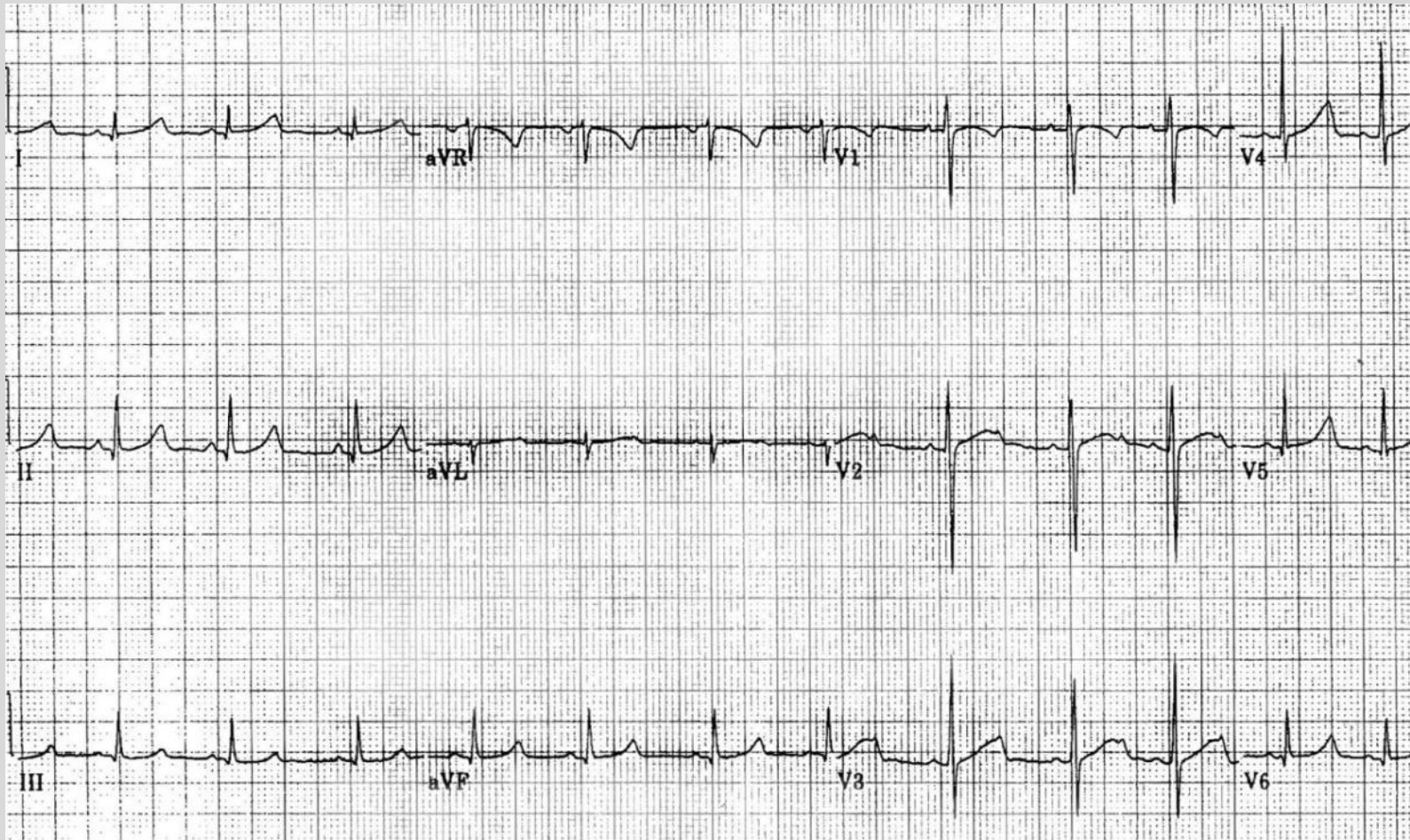
- Orthostatic hypotension
- psychogenic
- Long QT interval
- Vasovagal
- Hypoglycemia

What is the cause of her syncope?

- Orthostatic
hypotension****
- psychogenic
- Long QT interval
- Vasovagal
Hypoglycemia

DIAGNOSIS?

What if the ECG(EKG) looked like this?



- SYNCOPE = involuntary loss of posture & consciousness
-

- Many causes:
 - Vasovagal-interruption of normal central vasomotor tone leading to arteriolar vasodilation, hypotension, and decreased cerebral perfusion pressure. “Common faint” may be more common after fasting, associated w/anxiety, fright, surprise.
 - Vagovagal-when vagal stimulation causes extreme bradycardia, i.e., suctioning a newborn’s posterior pharynx
 - Cardiac-usu. associated w/heart lesions that cause obstruction to L-sided blood flow, ex: aortic stenosis
 - Arrhythmogenic-arrhythmias ex: long QT syndrome
 - Orthostatic- failure of vascular compensatory responses to postural changes leading to a decrease in BP and cerebral perfusion, ex: dehydration

Syncope-many causes (continued)

- Seizure disorders-noncardiac
- Metabolic-hypoglycemia
- Behavioral-hysterical(psychogenic pseudosyncope),
breath holding, hyperventilation

LONG QT SYNDROME- Arrhythmogenic cause of syncope

- Myocardial repolarization disorder
- May progress to ventricular arrhythmias(torsades de pointes)- then syncope
- May lead to sudden death
- QT interval usu. $>.46$ sec
- May be congenital or acquired, F>M
- Congenital: Romano-Ward Syndrome(nl hearing, autosomal dominant), Jervell and Lange-Nielsen Syndrome(deafness, autosomal recessive), Anderson-Tawil Syndrome(autosomal dominant), Timothy Syndrome(sporadic)

Ion channel disturbances, mutations in the genes responsible for Ca^{+} , K^{+} , and Na^{+} are implicated in congenital LQTS

- Acquired: drugs, electrolyte disturbances, underlying disease like hypothyroidism
- May have history or FHx of congenital deafness, +/-or sudden death

Syncope in Children (work-up)

- Most are vasovagal or noncardiogenic, last less than 1 minute
- Use history to your advantage:
 - Incontinence: prob seizure (non-cardiac)
 - Prior cardiac surgery: prob arrhythmia
 - Family history of sudden cardiac death/congenital deafness: prob arrhythmia
 - Exercise-induced: prob hypertrophic obstructive cardiomyopathy or arrhythmia
 - Longer duration: prob structural cardiac or arrhythmia
 - Witnessed cardiac arrest: probably not neurologic

Syncope- Treatment

- Treat the underlying cause, when applicable
- Reassurance, increase fluid intake, increase salt intake

Case #2

CC: 7 yo female
with a red
swollen knee

HPI-case #2

A 7 yo female presents for her 1st visit to the pediatrician's office since arriving in the US.

3 weeks ago, she presented to Border Control in Texas because of a 2-day history of fever with a Tmax of 104°F oral (40°C) and a sore throat. Symptomatic treatment recommended.

Physical exam in Texas revealed:

- Enlarged, slightly tender b/l anterior cervical lymph nodes
- Oropharyngeal injection with soft palate petechiae
- Rapid strep. test neg, no throat culture sent
- CVR: increased rate, nl rhythm. Lungs CTA no RRW
- Skin: no rash

Today, she presents with a 1-week history of intermittent/fleeting and random knee and wrist swelling accompanied by pain and redness. Decreased activity, tired, and developed a flat pink arm rash with a pale center 2 days ago.

Case #2-HPI continued

- All: none
- PSHx: none
- PMHx: noncontributory
other than HPI
- Meds: APAP for pain

- VS: HR 130, RR 30 BP 110/70, T 100.8F, Pulse ox 97%
- Alert, WH, uncomfortable, quiet
- HEENT: noncontributory, post. pharynx normal, soft palate normal
- CVR: increased HR, gr 2/6 holosystolic murmur best heard at the apex, lungs: CTA no RRW
- Abd: noncontributory
- Ext: R knee-warm, tender, erythematous, swollen, decreased range of motion secondary to pain
- Neuro: no obvious abnormalities

PHYSICAL EXAM- case #2

- CRP: +
- ESR: elevated
- Antistreptolysin O titer: elevated
- WBC count: elevated
- EKG: first degree heart block
- CXR: mild cardiomegaly

LABS for Case #2

- A. Kawasaki disease
- B. Acute rheumatic fever
- C. Lupus
- D. Cardiomyopathy
- E. Juvenile idiopathic arthritis

DIAGNOSIS?

- A. Kawasaki Disease
- B. Acute rheumatic fever***
- C. Lupus
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DIAGNOSIS?

Acute Rheumatic Fever (ARF)

Rheumatic Fever (Treatment)

- Best treatment = prevention
 - Antibiotics used for strep pharyngitis is for **ARF prevention**, not treatment for the pharyngitis
- Once diagnosed with ARF, long-term penicillin prophylaxis
 - Oral twice daily
 - IM once monthly
- Treatment of ARF = eradication of strep using penicillin, plus NSAIDs +/- steroids
- Treatment of CHF / valvular disease as necessary
- NSAIDs may mask migratory polyarthrititis, best to use APAP prior to diagnosis

CASE #3

- CC: 6-month-old male presents with irritability, sweating, decreased feeding, and rapid breathing x 2 days. 1 week ago, he had a body rash following a low-grade fever for 2 days. He was otherwise well.
- Birth Hx: FT AGA via SVD, uncomplicated pregnancy and birth
- Meds: none
- PMHx/PSHx: noncontributory

- Inconsolable, pale, WD, WH, sweating, ant. fontanelle open and flat,
- VS: T 98.5F, HR 180, RR 70, pulse ox 92%
- HEENT: unremarkable
- Neck: supple
- CVR: increased HR, diminished sounds, S3 present. Increased resp rate, wheezing throughout, positive intercostal retractions
- Abd: positive hepatosplenomegaly, positive BS
- Ext: capillary refill 3 seconds, diminished peripheral pulses
- Skin: no rash, pale

Physical exam- case #3

- EKG: low voltage QRS, ST-segment depression
- CXR: cardiomegaly, pulmonary edema, elevated diaphragms

LABS/IMAGING

- a. Kawasaki disease
- b. Acute rheumatic fever
- c. Truncus arteriosus
- d. Hypoplastic left heart syndrome
- e. Dilated cardiomyopathy

DIAGNOSIS?

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DIAGNOSIS

Cardiomyopathy-disease of the myocardium

Myocardial Diseases: based on etiology

A. Infectious-Viral, nonviral, immunologic(Kawasaki syndrome, acute rheumatic fever)

B. Metabolic-

1.Electrolyte-
hypoglycemia,magnesemia,kalemia and hyperkalemia

2.Endocrine-
hypothyroid,pheochromocytoma, infant of diabetic mother

3.Drugs-
etoh,cocaine,radiation,chloroquine,adriamycin

4. Vitamin/trace metal deficiencies-
thiamine, selenium, carnitine, taurine

C. Infiltrative-Glycogen storage/glycolipid diseases, hemochromatosis, amyloidosis, sarcoidosis, neoplasms, mucopolysaccharidoses

D. Ischemic- rare, congenital anomalies of coronary arteries

Cardiomyopathy: Disease of the myocardium: usu have a large heart(more muscle or larger cavity)

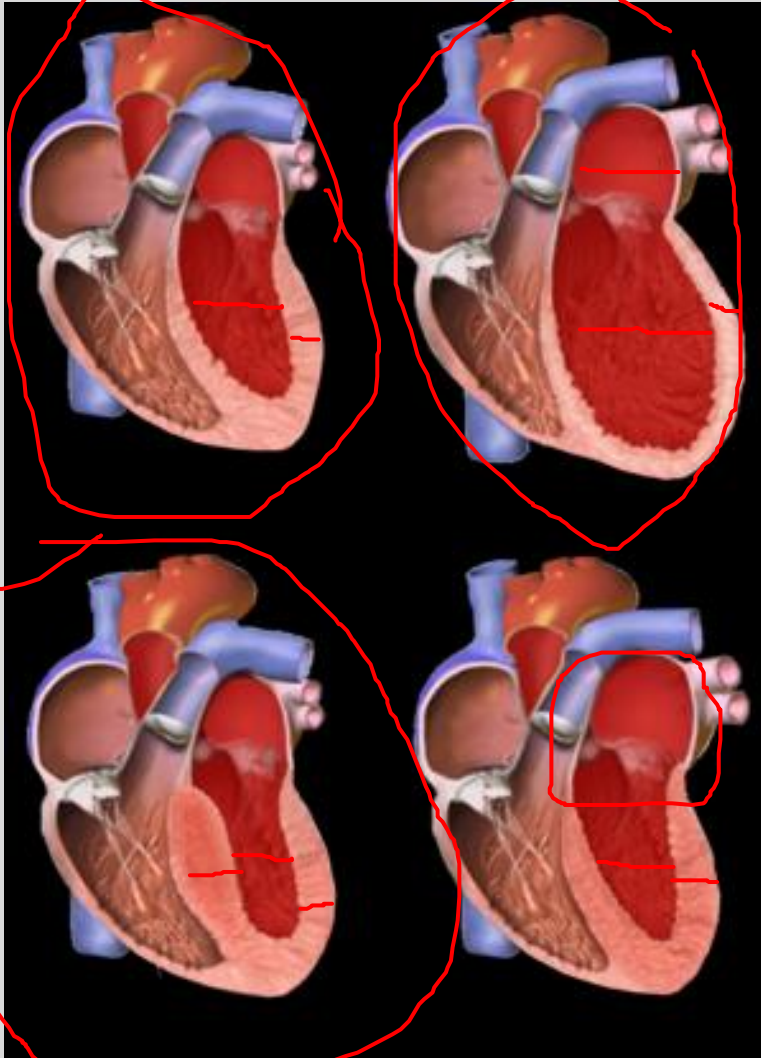
Primary Myopathy Classification:

- DILATED-most common, may be idiopathic, infectious (Coxsackie B virus or echovirus), postinfectious, familial, secondary to systemic disease or cardiotoxic drugs
- HYPERTROPHIC- familial 50% of patients
- RESTRICTIVE-rare in pediatrics, idiopathic, secondary to systemic disease

PRIMARY MYOPATHIC

Based on Physiology:

- **Dilated:** systolic dysfunction
- **Hypertrophic:** diastolic dysfunction
- **Restrictive:** diastolic dysfunction



Dilated Cardiomyopathy-most common

- Myocyte atrophy increases left ventricular(LV) cavity, decreases contractility
 - Low ejection fraction(EF)- Large LV with decreased pump ability
 - Pumping defect = systolic dysfunction
- Presents with CHF
 - Decreased cardiac output
 - May present with LBBB
- Identify cause, if possible, to help treatment-but often idiopathic
 - Reversible (etoh, infection, pregnancy, malnutrition)
 - Irreversible (doxorubicin, genetic)
- Treatment:
 - Treat causative agent (if possible) and symptoms(CHF, heart block)
 - Heart transplant
- Prognosis: 50% mortality within 5 years

Hypertrophic Cardiomyopathy(HCM)

- Myocyte hypertrophy decreases LV cavity size
 - High EF (LV cannot fill, but pumps it all out)
 - Filling defect = Diastolic dysfunction
- Asymptomatic until sudden cardiac death
 - Increased HR (running / sports) can cause arrhythmia in pathologic myocardium
- Treatment: Symptomatic and
 - Myomectomy-resection of subaortic muscle may help symptoms
 - Heart transplant
- Prognosis: sudden death 5% pts per year

Restrictive Cardiomyopathy-Rare

- Diastolic dysfunction(extremely poor compliance of the ventricular walls)
- Ventricular volume is normal to mildly decreased
- Wall thickness is normal to minimally increased
- Typically, atrial enlargement
- Treatment: control symptoms and maintain BP control without decreasing cardiac output, cardiac transplant
- Prognosis: poor, 5-year survival <30%



CC: 18-month-old
male with fever
for 6 days

Case #4

HPI case #4

An 18-month-old male presents to the pediatric office with 6 days of fever, T_{max} of 104 °F axillary(40 °C). Mom states he has red eyes and lips, a rash, swollen hands and a swollen gland in his neck.

Case #4 Physical Exam

- VS: 104.4F, HR 160, RR 30, pulse ox 96%
- Nontoxic appearing, crying but consolable in mom's arms, WH, hot to touch, WD
- Eyes: conjunctiva- injected b/l, no discharge, PERRLA, EOMI
- Mouth: no oral ulcers, + injection of the oropharynx , tongue with red, prominent papillae, lips-cracked and erythematous
- Ears: TM's pink, flat, positive light reflex
- Neck: supple, 2 x 2cm right tonsillar lymph node, mobile, nontender
- CVR: increased HR, no murmur, lungs: CTA no RRW
- Ext: hands/feet: nonpitting edema
- Rapid strep. test neg. in office, throat culture is pending

- CBC: elevated WBC's, low Hgb/Hct, plt normal
- ESR: elevated
- UA: WBC's, no bacteria, + protein
- ECG: sinus tachycardia

LABS

- a. Scarlet fever
- b. Stevens-Johnson syndrome
- c. Juvenile idiopathic arthritis
- d. Kawasaki disease
- e. Acute rheumatic fever

DIAGNOSIS?

- a. Scarlet fever
- b. Stevens-Johnson syndrome
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- d. Kawasaki disease***
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DIAGNOSIS

Kawasaki Disease (mucocutaneous lymph node syndrome)

Kawasaki Disease KD (Epidemiology)

- Generalized vasculitis of unknown etiology; ?mimicry of an infectious agent
- Has endemic & epidemic patterns: most common in winter/spring
- More common in children (80% < 8 years), more common in Asian ancestry
- A leading cause of acquired heart disease in children

Kawasaki Disease (Diagnosis)

- Fever >5 days PLUS 4 of the following:
 1. Changes in the extremities: Acute : erythema/edema of the hands/feet
Subacute: desquamation hands/feet
 2. Changes in the oral mucosa (include red/cracked lips, injected oral mucosa and pharynx, strawberry tongue)
 3. Polymorphous exanthema
 4. Bilateral conjunctival injection w/o pain or exudate
 5. Unilateral cervical lymphadenopathy
- Lab findings may include elevated WBC's, anemia, normal plt count early and thrombocytosis late
- Coronary aneurysms are a major concern in untreated disease

Strawberry Tongue

- KD causes white exudate on the tongue. Papillae enlarge due to erythema & inflammation
- Appearance changes as white exudate disappears along with desquamation of the tongue surface (with a decrease in sense of taste)
- Causes of strawberry tongue include (from most to least common):
 - Scarlet fever (group A beta-hemolytic strep with enterotoxin)
 - Kawasaki disease
 - Toxic-like shock syndrome



Source: Knoop KJ, Stack LB, Storrow AB, Thurman RJ: *The Atlas of Emergency Medicine*, 3rd Edition: <http://www.accessmedicine.com>
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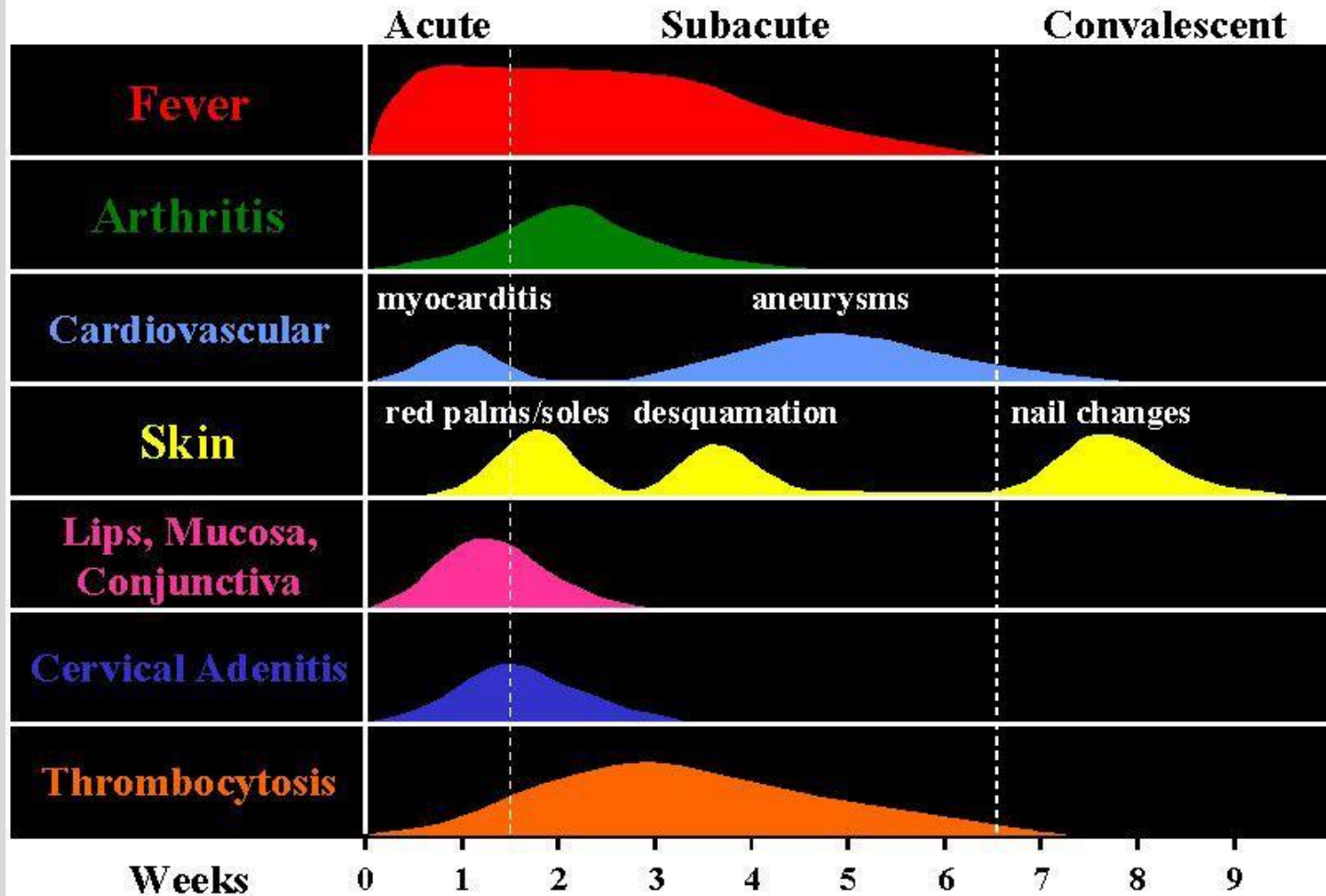


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Kawasaki Disease (Symptoms)

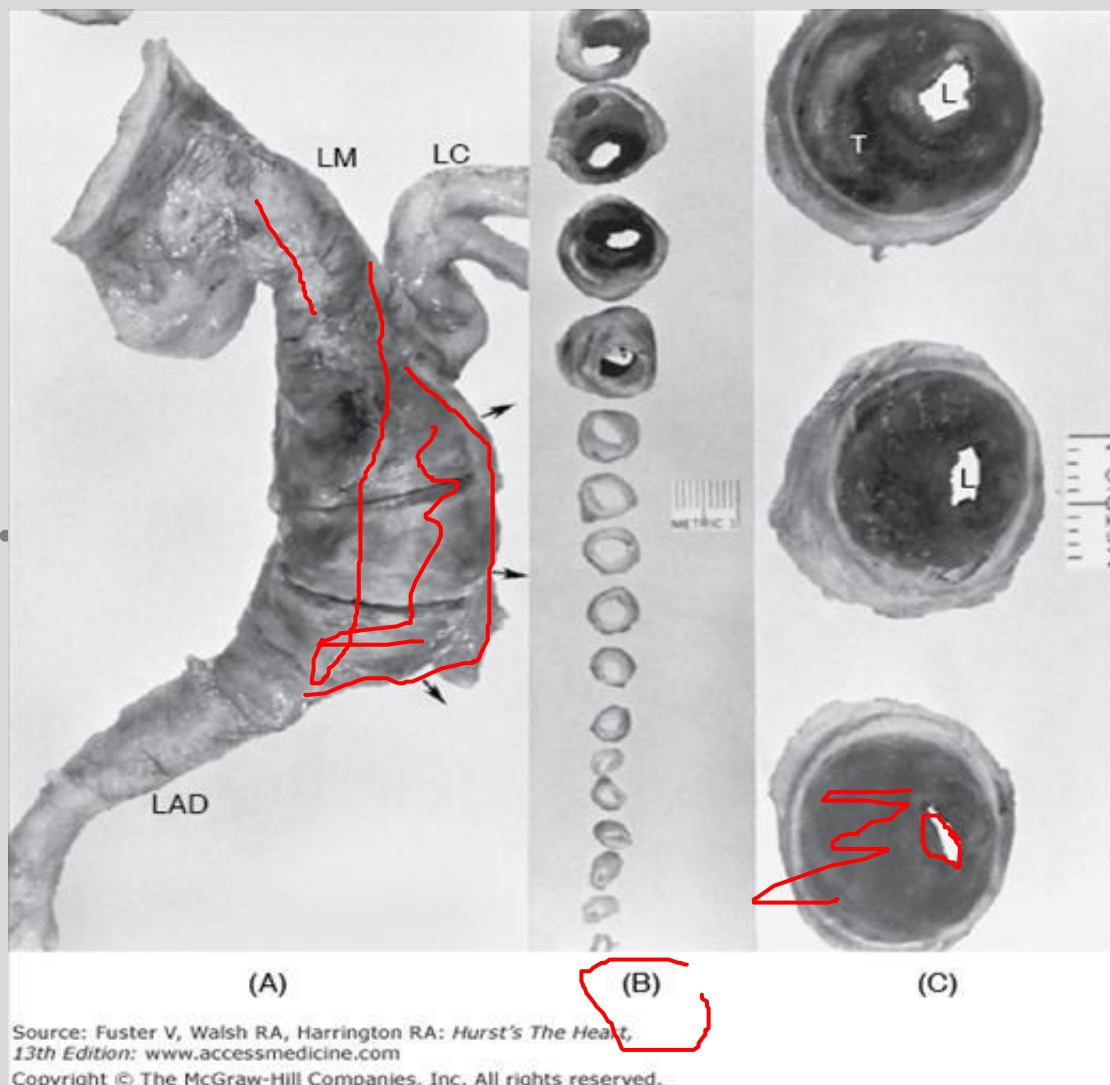
- Disease has 3 phases:
 - * 1. Acute Febrile –usu. 1- 2 wks. of illness(unremitting fever lasts usu 1-2 weeks but may persist up to a month), necrotizing arteritis, saccular aneurysms may form
 - ** 2. Sub-acute- usu lasts 3-4 weeks(subacute/chronic vasculitis which may last years), desquamation, thrombocytosis
 - ** 3. Chronic or “convalescent”-lasts up to 2 mos. Usu resolves 6-8wks after onset. Vessels of vasculitis develop smooth muscle myofibroblasts leading to stenosis
- Most symptoms(*) are apparent in acute phase, more serious sequelae(**) occur in sub-acute or chronic phase
- Thus, early diagnosis and treatment is of the utmost importance

Clinical Manifestations of Kawasaki Disease



Kawasaki Disease-untreated (Coronary artery disease stages)

- Stage 1: (first 2 wks.) acute **vasculitis** of coronary arteries.
- Stage 2: (2-4 wks. after onset) vasculitis persists, may lead to a weakening of the wall and **aneurysms** may occur. Flow decreases with the increasing diameter. Stagnant blood in aneurysm may lead to thrombosis. Thrombus in coronary arteries may lead to obstruction of coronary blood flow.
- Stage 3: (4-8wks after onset) artery inflammation subsides but aneurysms may persist.
- Stage 4: (>8wks after onset) scar formation and **calcification** of the arteries with stenosis and recanalization of the lumen, myocardial fibrosis.
- 20% of untreated children will develop coronary artery disease.



A: Left coronary artery aneurysm

B: Cross sections taken throughout length of same vessel

C: Magnified images from center clearly showing CA stenosis;
T=thrombus
L=lumen

Kawasaki Disease (Treatment)

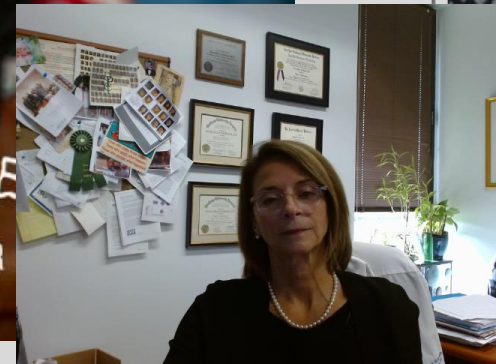
- Initial therapy: Reduce inflammation and prevent coronary aneurysms!
- Chronic therapy: Prevent coronary artery thrombosis
- IVIG (intravenous immunoglobulin): first-line tx'ment to decrease immune response-decreases risk of coronary artery aneurysm if used in the first 10 days of symptoms. Approx. 85% success rate.
- ACUTE(anti-inflammatory): Aspirin 80-100mg/kg/d divided QID: reduces systemic symptoms and decreases risk of aneurysm formation
- CHRONIC(antithrombotic): Aspirin 3-5mg/kg qD when afebrile and discontinued 6-8wks after the onset of illness UNLESS coronary aneurysms are present(continue low dose aspirin indefinitely)
- Second dose IVIG or steroids or TNF-alpha antibody are used in more resistant cases.

- **Syncope**: Must identify cause to prevent subsequent episodes
- **Cardiomyopathy**: Intrinsic disease of the heart muscle
- **ARF**: Strep pharyngitis sequela; JONES criteria for diagnosis; damages valves; requires long-term prophylaxis to prevent recurrence
- **Kawasaki disease**: Vasculitis; clinical diagnosis; treatment focuses on coronary aneurysm prevention and symptomatic treatment

Thank you

“IF IT DOESN'T MAKE THE WORLD BETTER
- DON'T DO IT.”

KID PRE
KID PR



<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>